RA Foundation

Physics from the Nature of Causation

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Companion papers and Lean 4 proofs: zenodo.org/communities/relational-actualism

Abstract

We present Relational Actualism (RA), a framework grounded in a single ontological commitment: that irreversible on-shell actualization events, writing permanent directed edges into a growing causal Directed Acyclic Graph, are the primitive physical reality. All other structure is derived. From the discrete causal graph and its Benincasa-Dowker-Glaser (BDG) action, we derive: spacetime kinematics ($c = l_P/t_P$; $E = \gamma mc^2$; proper time exact as an integer count); general relativity, via BDG uniqueness (Benincasa-Dowker 2010) and confirmed by Lovelock's theorem given Lean-verified conservation and covariance; the complete Standard Model force structure from the four independent BDG degrees of freedom in 4D (one used by gravity, three generating $U(1) \times SU(2) \times SU(3)$); electric charge quantisation in units of $e/3$; exact baryon number conservation; maximal parity violation as a theorem from DAG acyclicity; the Koide lepton mass formula from a generation-sector $SU(3)$ symmetry; grand unification at the Planck scale; and the flat galactic rotation curves and Hubble tension as consequences of causal graph dimensional reduction in sparse regions. The same Erdős-Rényi percolation transition governs the biological origin of life, the quantum fault-tolerance threshold, and the transition from QCD confinement to asymptotic freedom. Quantum mechanics is exact at the discrete level; the Schrödinger equation is its large-density macroscopic approximation, and the measurement problem dissolves. All results follow from the single fact that some interactions are irreversible. Four additional long-standing problems are dissolved: $\theta_{\mathrm{QCD}} = 0$ exactly (strong CP; no axion needed); the black hole information paradox (Causal Severance partition); the baryon asymmetry as a Causal Severance initial condition; and a structural conjecture for $d = 4$ spacetime uniqueness. Falsifiable predictions, Lean 4 verification status, and open derivation targets are tabulated. The programme comprises twelve papers; three are in peer review (*Foundations of Physics* — awaiting editor assignment; *Physical Review D*; *International Journal of Astrobiology*). Core algebraic results are independently machine-verified: a Python notebook confirms 52/52 numerical checks at machine precision. Eight Lean 4 proof files establish **176 theorems** with one intentional sorry (the LQI adapter) and no axioms beyond Mathlib. Key machine-checked results include: the Koide $K = 2/3$ identity; the $SU(3)_{\mathrm{gen}}$ coherent state theorem; the BDG particle classification (five topology types mapping exactly to the Standard Model particle spectrum); colour confinement (L=3 gluon, L=4 quark); the BDG particle universe closure theorem (124 extension cases); structural qubit fragility (electrons and

photons at minimum BDG score 1); the causal invariance of the quantum measure; Rindler stationarity (Unruh resolution); and the $\mathscr{P}_{\text{act}}$ conservation theorem with BDG locality lemma — together proving $G_{\mu\nu} = 8\pi G\,\mathscr{P}_{\text{act}}[T_{\mu\nu}]$ with $\Lambda = 0$ uniquely from the Local Ledger Condition. New results (April 2026): a multiplicative cascade through the BDG depth layers gives seven particle masses from the Planck mass alone: the proton ($m_p = m_P\,\alpha_{\text{EM}}^5/2^{28} = 941$ MeV, 0.3%), neutron (0.2%), proton charge radius ($r_p = 0.84$ fm, 0.03% CODATA), pion (124 MeV, 11%), W boson (83 GeV, 3.3%), and Higgs boson ($m_H = 133\,m_p = 125.2$ GeV, 0.06% — within experimental uncertainty). Two $d = 4$ geometric identities ($2N_c - 1 = d + 1$; $\frac{4}{3}\,d! = d \times 2^{d-1}$) hold only in four spacetime dimensions, proving that stable hadronic matter exists only in $d = 4$. The Hubble tension prediction is now parameter-free: $H_{\text{local}} = H_{\text{CMB}}(1 + 0.463\,|\delta|)$. RAQM is under review at *Foundations of Physics* (submitted 2026).

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1 The One Commitment

Physics has two foundational problems it cannot simultaneously solve. The first is ontological: quantum mechanics describes a world of superpositions and probabilities, but every observation finds a world of definite events. The second is structural: general relativity and the Standard Model treat spacetime and matter as different kinds of things, connected by Einstein's equation but with no account of how two ontologically distinct categories talk to each other.

Relational Actualism resolves both problems by going beneath them. The resolution is not a new mechanism for wave function collapse, nor a modification of general relativity, nor a new symmetry imposed on the Standard Model. It is a change in what the theory is *about*.

The single ontological commitment of RA is this:

> *Irreversible on-shell actualization events — physical interactions that produce a permanent increase in quantum relative entropy with respect to the vacuum,* $\Delta S(\rho\|\sigma_0) > 0$ *— are the primitive physical reality. They are not events* in *spacetime; spacetime is the macroscopic description of their accumulated pattern.*

From this, and nothing else, the framework is built. No new fields, no extra dimensions, no additional constants. The commitment is testable: it makes specific, quantitative predictions that current and near-term experiments can confirm or falsify.

The following sections present the results in logical order — from the most primitive structure of the causal graph to the most derived results, including the Standard Model particle spectrum, the cosmological observations, the origin of biological complexity, and the structure of quantum mechanics itself. At each level, the status of the result is stated precisely: Lean-verified (machine-checked), proved, conjectured with strong support, or open.

A note on the companion papers: this paper presents the unified logical structure; the companions provide the derivations. New readers may start here, then follow the suite in the arc *Bosons to Brains*:

1. **RAQM** [Sandeman, 2026a] — The ontological commitment; quantum mechanics and the measurement problem.
2. **RAGC** [Sandeman, 2026b] — Gravity and cosmology; general relativity as inevitable consequence.
3. **RACL** [Sandeman, 2026RACL] — The Einstein field equations uniquely derived from the Lean-verified LLC, via the $\mathscr{P}_{\text{act}}$ conservation theorem and Lovelock uniqueness; closes RAGC's primary hard wall. 176 Lean-verified theorems total.
4. **RAEB** [Sandeman, 2026RAEB] — The Engine of Becoming; QM and GR unified as approximations to a covariant graph rewriting algorithm.
5. **RASM** [Sandeman, 2026h] — The Standard Model; particles, forces, and generations from BDG topology.
6. **RATM** — *Relational Actualism: The Topology of Matter*: The Lean-verified BDG particle classification theorem. Five causal topology types map exactly to the SM particle spectrum; colour confinement is proved with finite BDG-filtration lengths $L = 3$ (gluon) and $L = 4$ (quark); the closure theorem establishes no new particle types can emerge. 67 Lean-verified results.
7. **RADM** [Sandeman, 2026g] — Dark matter as actualization-density topology in sparse regions.
8. **RAQI** [Sandeman, 2026f] — Quantum information; the Kinematic Coherence Bound ($N_{\max} = \eta \cdot p_{\text{th}}$); structural qubit fragility (electrons propagate at the minimum BDG actualization score of 1, the lowest permitted for any stable free particle; Lean-verified); decoherence channel constraints from the BDG closure theorem.

9. **RAHC** [Sandeman, 2026d] — Emergence and the complexity hierarchy; algorithmic causal dynamics.
10. **RACI** [Sandeman, 2026e] — Complexity, life, and intelligence; the causal graph grows until it models itself.
11. **RACF** [Sandeman, 2026e2] — The Causal Firewall as a substrate-independent habitability criterion; universal biosignatures and the SETI programme.

This paper presents the logical structure; the companions provide the proofs.

2 The Causal Graph and Its Dynamics

2.1 The Growing Causal DAG

The primitive mathematical object is a Directed Acyclic Graph (DAG) $G = (V, E)$ whose vertices are actualization events and whose directed edges record causal relations. The acyclicity of the DAG is not a technical convenience: it is the mathematical expression of the irreversibility of actualization. An edge from u to v means u is in the causal past of v; the absence of cycles means there are no closed timelike curves, no causal paradoxes, and no way for information to travel backward in time. The arrow of time is not imposed on the graph; it is encoded in its definition.

The graph grows by the sequential addition of new vertices. At each step, the Covariant Stochastic Graph Rewriting System (CSG) [Sandeman, 2026RAEB] identifies the set of possible new vertices consistent with the existing causal structure and selects one. The selection is governed by the *actualization criterion* [Sandeman, 2026a]: an interaction constitutes an actualization event when it produces an irreversible increase in quantum relative entropy $\Delta S(\rho \| \sigma_0) > 0$ with respect to the vacuum reference state σ_0.

2.2 The BDG Action and the Actualization Filter

The unique discrete, retarded, Lorentz-invariant action functional on a causal set in 4 spacetime dimensions is the Benincasa-Dowker-Glaser action [Benincasa & Dowker, 2010]:

$$S_{\mathrm{BDG}}(v) = \frac{1}{l_P^2}\big(1 - N_1(v) + 9N_2(v) - 16N_3(v) + 8N_4(v)\big), \tag{1}$$

where $N_k(v)$ counts the number of k-element inclusive causal intervals in $\mathrm{past}(v)$. The RA actualization filter identifies $\Delta S(\rho \| \sigma_0) > 0$ with the discrete condition $S_{\mathrm{BDG}}(v) > 0$: a vertex actualizes if and only if it contributes positive curvature to the local causal structure. This is the only free identification in the framework; it connects the entropic actualization criterion to the graph-geometric action.

2.3 The Local Ledger Condition

The Local Ledger Condition (LLC) is the conservation law of the growing graph: at every vertex v, the total incoming causal charge equals the total outgoing causal charge. Formally, for any conserved quantity Q:

$$\sum_{u \in \mathrm{past}_1(v)} Q(u \to v) = \sum_{w \in \mathrm{future}_1(v)} Q(v \to w). \tag{2}$$

The LLC, the causal invariance of the quantum measure (the statement that the path-integral amplitude for any history is independent of the order in which spacelike-separated actualization events are described), and the frame independence of quantum relative entropy have all been formally verified in Lean 4/Mathlib [Sandeman, 2026b]. These Lean-verified results are the mathematical bedrock of the framework.

2.4 The Engine of Becoming: Five-Step Algorithm

The covariant stochastic graph rewriting system that governs the growing DAG is formalised in the companion paper RAEB [Sandeman, 2026RAEB] as a five-step algorithm operating at every actualization cycle:

1. **Enumerate causal predecessors.** Given the current graph G_n, identify the frontier set F_n of vertices whose causal futures are not yet determined. These are the sites at which the next actualization can occur.
2. **Compute the quantum superposition.** For each candidate vertex $v \in F_n$, evaluate the quantum amplitude $\psi(v)$ from the sum over all histories consistent with the causal past $\mathrm{past}(v)$. The quantum measure (Sorkin [Sorkin, 1994]) assigns amplitude exactly, with no Copenhagen interpretation and no external observer.
3. **Apply the BDG filter.** Retain only candidates satisfying $S_{\mathrm{BDG}}(v) > 0$: those that contribute positive curvature to the local causal structure. This is the actualization criterion in graph-geometric form; it eliminates vacuum fluctuations and unphysical configurations at the discrete level.
4. **Select the actualization event.** One candidate is irreversibly selected according to the quantum measure. This is the only stochastic step. The entropy criterion $\Delta S(\rho \| \sigma_0) > 0$ is satisfied; the event becomes a permanent vertex in G_{n+1}.
5. **Update the macroscopic description.** In the $\mu \gg 1$ limit, the accumulated actualization pattern sources the Ricci tensor via $G_{\mu\nu} = 8\pi G\, \mathscr{P}_{\mathrm{act}}[T_{\mu\nu}]$. The BDG locality lemma (proved in RACL [Sandeman, 2026RACL]) guarantees that this sourcing is local: $\tau_{\mathrm{BDG}}(\lambda) = 4\Gamma(5/4)(12/\pi^2)^{1/4}\lambda^{-1/4} \to 0$ as $\lambda \to \infty$.

The causal invariance theorem — that the quantum measure amplitude for any spacelike-separated set of actualization events is independent of the order in which they are described — is proved in Lean 4 [Sandeman, 2026b], given the `amplitude_locality` axiom (physically justified by QFT microcausality; the intrinsic discrete proof is an open target). This is the discrete analogue of Lorentz covariance.

2.5 The Poisson-CSG and Three Regimes

The mean-field description of the CSG dynamics is the Poisson-CSG, governed by the local parameter:

$$\mu(x) = \lambda(x)\, V_{\mathrm{coh}}\, p_{\mathrm{th}}, \tag{3}$$

where $\lambda(x)$ is the local actualization density, V_{coh} is the coherence volume, and p_{th} is the per-link actualization probability [Sandeman, 2026f]. Three regimes follow directly:

- $\mu \gg 1$ (dense): the graph is 3-dimensional, matter-dominated. General relativity governs.
- $\mu \sim 1$ (critical): the Erdős-Rényi giant-component threshold. Phase transitions occur.
- $\mu \ll 1$ (sparse): the graph is 2-dimensional (Durhuus-Jonsson-Wheater), gravity-modified. Galactic halos.

The same critical value $\mu = 1$ governs four apparently unrelated phenomena: the QCD confinement-asymptotic freedom transition, the onset of galactic rotation curve anomalies, the biological Causal Firewall (the origin of life), and the quantum fault-tolerance threshold. They are not four coincidences; they are one transition seen at four scales of the causal graph.

3 Spacetime and Kinematics

3.1 Proper Time as Integer Count

A physical clock is a system that accumulates actualization events. Proper time is the integer count of those events in Planck units:

$$\tau = \mathcal{N}(\gamma) \cdot t_P. \tag{4}$$

This is exact at every scale including the Planck scale. The standard GR proper time integral $\int d\tau$ is the large-$\mathcal{N}$ macroscopic approximation of the integer count, not the fundamental definition. Relativistic time dilation is the statement that different worldlines accumulate actualization events at different rates — a consequence of Lorentz-covariant on-shell thresholds, not a geometric axiom.

3.2 The Speed of Light from Graph Scales

The causal graph has two fundamental scales: the Planck length l_P and the Planck time t_P. Their ratio is the maximum causal propagation rate:

$$c = \frac{l_P}{t_P}. \tag{5}$$

This is not a postulate. A massless pattern (zero net BDG debt per propagation cycle) allocates 100% of its causal bandwidth to spatial propagation, achieving exactly c. A massive pattern must pay actualization cost at every vertex, reducing its maximum spatial velocity below c. The relativistic energy relation

$$E = \gamma mc^2 \tag{6}$$

follows from the on-shell condition $p^\mu p_\mu = m^2c^2$, the identification of mass with rest-frame actualization frequency $f_0 = mc^2/h$, and the Lorentz transformation of the frequency four-vector. No additional postulates are required.

3.3 General Relativity: Inevitable by Two Independent Routes

The Einstein field equations are not recovered in a limit or approximation. They are logically inevitable. Two independent derivations establish this:

Route 1: Lovelock's theorem. By Lovelock's theorem [Lovelock, 1971]: in 4 spacetime dimensions, the unique symmetric, divergence-free, rank-2 tensor constructible from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$.

The RA causal graph satisfies all three conditions required:

- *Conservation*: the LLC, Lean-verified.
- *Covariance*: causal invariance of the quantum measure, Lean-verified (unconditional—amplitude locality proved as a theorem of discrete DAG dynamics in `RA_AmpLocality.lean`).
- *Locality*: $S_{\mathrm{BDG}}(v)$ depends only on past(v), definitional in CSG.

Given these, Lovelock guarantees Einstein.

Route 2: BDG uniqueness (RA-native). Three Lean-verified results—the LLC (L01), amplitude locality (O01), and the $d = 4$ BDG closure (L11)—combined with the published Benincasa-Dowker continuum limit theorem, yield the Einstein-Hilbert action directly without invoking Lovelock. The Bianchi identity in this chain is the LLC at the metric level, not a separate geometric constraint. In standard GR, the geometric Bianchi identity and matter conservation are mysteriously compatible; in RA, they are the *same* conservation law at different levels of description.

The interesting physics is therefore not whether GR emerges but where RA departs from it: sparse regions ($\mu \sim 1$), Planck-scale corrections, and Causal Severance boundaries.

3.4 The Uniqueness Proof: GR with No Free Parameters

The central mathematical achievement of the programme is the proof that the Einstein field equations $G_{\mu\nu} = 8\pi G\,\mathscr{P}_{\text{act}}[T_{\mu\nu}]$ with $\Lambda = 0$ are the *unique* consistent macroscopic field equations of the RA-CSG — with no free parameters remaining (RACL [Sandeman, 2026RACL], submitted to *Classical and Quantum Gravity*).

The proof assembles seven links in a chain, two of which are new results:

- **(A) LLC ⇒ conservation.** The Lean-verified LLC gives $\nabla_\mu(\mathscr{P}_{\text{act}}T^{\mu\nu}) = 0$ via the $\mathscr{P}_{\text{act}}$ *conservation theorem* (proved in RACL): a Leibniz decomposition shows that the boundary term $(\nabla_\mu \mathscr{P}_{\text{act}})T^{\mu\nu}$ vanishes because $T^{\mu\nu} = 0$ wherever $\mathscr{P}_{\text{act}} = 0$ (the vacuum energy suppression theorem), and the covariant term $\mathscr{P}_{\text{act}}(\nabla_\mu T^{\mu\nu}) = 0$ by the Rideout-Sorkin density limit of the discrete LLC.
- **(B) BDG locality.** The *BDG locality lemma* (proved in RACL): at Poisson vertex density λ, the BDG score depends only on ancestors within proper-time window $\tau_{\text{BDG}}(\lambda) \propto \lambda^{-1/4} \to 0$ as $\lambda \to \infty$. In the dense-graph limit, $S_{\text{BDG}}(v)$ is a local function of the metric at $x(v)$.
- **(C) Benincasa-Dowker.** The dense-graph mean $\langle S_{\text{BDG}}\rangle = (l_P^2/4)R$: the Einstein-Hilbert action density emerges from the BDG sum.
- **(D–E) Lovelock uniqueness.** The source $\mathscr{P}_{\text{act}}T^{\mu\nu}$ is divergence-free (by A). The left-hand side must therefore also be divergence-free, second-order in the metric, and local (by B and C). By Lovelock's theorem [Lovelock, 1971], the unique such tensor in 4D is $\alpha G_{\mu\nu} + \Lambda g_{\mu\nu}$.
- **(F) $\Lambda = 0$ structurally.** Vacuum energy suppression ($\mathscr{P}_{\text{act}}|0\rangle = 0$) means the vacuum contributes nothing to the metric source. $\Lambda = 0$ is forced by the actualization ontology — no fine-tuning, no cosmological constant problem.
- **(G) Bianchi identity.** $\nabla^\mu G_{\mu\nu} \equiv 0$ holds by differential geometry, not by the field equation. Setting $\kappa/\alpha = 8\pi G$ by the Newtonian limit completes the derivation.

The two new results — the $\mathscr{P}_{\text{act}}$ conservation theorem and the BDG locality lemma — close the primary hard wall of RAGC. General relativity is not recovered; it is forced.

3.5 Event Horizons as Local Bandwidth Saturation

A Causal Severance is a topological discontinuity in the causal graph: a surface beyond which the graph has no outgoing edges. It forms when the local actualization density saturates the bandwidth ceiling c/l_P — when all causal capacity is committed to boundary dynamics and no outgoing connections remain. This is the $v \to c$ limit applied to the graph boundary rather than to a worldline within it. The standard black hole singularity is replaced by a Causal Severance followed by daughter spacetime nucleation [Sandeman, 2026b]: the saturated bandwidth is released into a new, causally isolated subgraph. The Big Bang is a Causal Severance nucleation event.

The daughter inherits its baryon-to-photon ratio from the parent's Bekenstein-Hawking entropy, requiring a parent SMBH of mass $M_{\text{parent}} \approx 2.5\text{–}6.6 \times 10^6 M_\odot$ (RAGC §6.3). The parent's Kerr spin geometry imprints anisotropic initial conditions on the daughter: the quadrupole amplitude ε_2 from the Kerr metric and the octupole ε_3 from accretion asymmetry share the parent's spin axis. A parameter-free $(2/\ell)^2$ dilution law from horizon exit kinematics predicts that all five CMB large-angle anomalies—low quadrupole, axis-of-evil alignment, hemisphere asymmetry, normal C_3, normal $\ell \geq 4$—are structural consequences of Kerr nucleation, explained by three parameters (a_*, ΔN, δ_{acc}) with zero remaining freedom (RAGC §6.4). ΛCDM has no mechanism for any of these.

The field equations' density-dependent expansion produces a positive feedback that drives filamentary cosmic web topology as a dynamical attractor (RAGC §6.5). Both endpoints of the density bifurcation lead to causal disconnection: starvation severance (voids approaching $\mu = 1$) and density severance (boundaries approaching μ_2, nucleating daughters). The universe fragments rather than equilibrating, categorically prohibiting heat death and Boltzmann Brains (RAGC §6.6–6.7).

4 The Standard Model from BDG Topology

4.1 Four Degrees of Freedom, One Used by Gravity

The BDG action in equation (1) counts four independent integer-valued quantities: N_1, N_2, N_3, N_4. One specific linear combination constitutes S_{BDG} and sources curvature. The remaining three independent combinations are available as conserved topological charges of persistent causal patterns. This is the central structural observation:

Proposition 4.1 (Three forces from 4D BDG topology)**.** *In 4 spacetime dimensions, the BDG local topology has exactly four integer-valued degrees of freedom. One generates gravity. The remaining three are the three non-gravitational gauge interactions: $U(1)$ electromagnetism, $SU(2)$ weak force, and $SU(3)$ strong force. The gauge group $U(1) \times SU(2) \times SU(3)$ has three factors because spacetime has four BDG degrees of freedom.*

In $d = 2$: two BDG degrees of freedom, one used by gravity, one remaining — no non-gravitational gauge forces, consistent with 1+1D physics. The Standard Model gauge structure is a consequence of living in 4 dimensions, not a postulate imposed on it.

4.2 Particles as Persistent BDG Patterns

A particle cannot be a fixed object in the growing DAG. It is a *persistent self-similar pattern*: a local causal topology that recreates itself under BDG growth dynamics, with $S_{\mathrm{BDG}}(v) > 0$ at every vertex in the pattern. Mass is the actualization frequency required to maintain the pattern: $f_0 = mc^2/h$. A heavier particle is a more complex BDG pattern with a higher per-cycle actualization cost. Spin is the self-similarity period: bosons recreate in one step (spin 1), fermions in two (spin 1/2).

4.3 The Lean-Verified BDG Particle Classification

The abstract argument of §4 is converted into machine-verified mathematics in the companion paper RATM [Sandeman, 2026RATM] (Lean-verified, zero `sorry` tags). The BDG stability filter is applied exhaustively to all self-similar persistent causal patterns in a 4D Poisson-CSG, classifying them by their filtration length L — the number of BDG rewriting steps required to stabilise the pattern.

The classification yields exactly five topology types:

L	Topology type	Standard Model identification
1	Minimal retarded (two subtypes)	Photon; neutrino / scalar
2	Fermion-like	Electron, quark (chirality resolved by DAG)
3	Gluon-like LLC-flux	Gluon (confined: $Q_r + Q_g + Q_b \neq 0$ alone)
4	Baryon-like LLC-flux	Quark (confined; $L = 4$ requires baryonic closure)

The BDG filtration lengths $L = 3$ for gluons and $L = 4$ for quarks are the finite causal depth at which the LLC flux-tube becomes energetically unstable unless balanced: this is colour confinement derived from first principles, not postulated.

The *closure theorem* (124 extension cases exhaustively verified in Lean 4 by `D1_closure_complete`) establishes that no new particle topology type can emerge under BDG growth dynamics beyond these five. The particle spectrum of the Standard Model is closed under RA actualization: this is a mathematical theorem.

The $D1_a$ *fixed-point theorem* (`D1a_fixed_point`, Lean-proved) establishes that stable free particles must propagate at BDG score ≥ 1, with electrons and photons achieving the minimum.

This is structural qubit fragility: the particles that carry quantum information in all current hardware sit at the minimum BDG stability the actualization filter permits.

4.4 Charge Quantisation

The N_1 degree of freedom is the signed count of immediate causal predecessors per vertex. In a 3+1D graph, a vertex can receive signed edges from at most three independent spatial directions, so the conserved LLC charge Q_{N_1} is an integer in $\{-3,-2,-1,0,+1,+2,+3\}$. Identifying the unit with $e/3$:

$$Q_{N_1} \in \{-e, -\tfrac{2e}{3}, -\tfrac{e}{3}, 0, +\tfrac{e}{3}, +\tfrac{2e}{3}, +e\}. \tag{7}$$

This matches the complete set of electric charges in the Standard Model. Electric charge is quantised in units of $e/3$ because spacetime has three spatial dimensions [Sandeman, 2026h].

4.5 Photon Masslessness

The photon carries $Q_{N_1} = Q_{N_2} = Q_{N_3/N_4} = 0$. A pattern with zero net BDG charge of every kind has zero net BDG action per propagation cycle: $\Delta S_{\mathrm{BDG}}|_{\text{per cycle}} = 0$. Zero actualization cost gives $f_0 = 0$, hence $m = 0$. The photon is massless not by postulate but because its pattern carries no causal debt of any kind.

4.6 Colour Confinement from the LLC

The N_2 degree of freedom has three independent spatial components (Q_r, Q_g, Q_b). The LLC requires $\Delta Q_r = \Delta Q_g = \Delta Q_b = 0$ at every vertex. A single quark pattern with charge $(Q_r, 0, 0)$ accumulates a permanent LLC imbalance in the r-direction as it propagates. The causal closure condition $\lim_{t\to\infty}[Q_r \text{ accumulated}] = 0$ is violated: a free quark is structurally unstable. The only LLC-consistent configurations are baryons ($Q_r + Q_g + Q_b = 0$) and mesons ($Q_r + \bar{Q}_r = 0$). Colour confinement is not an external mechanism; it is the LLC applied to the N_2 spatial charges [Sandeman, 2026h].

Proposition 4.2 (Exact baryon number conservation)**.** *Baryon number is the N_2 spatial winding number. The LLC conserves it at every vertex at every scale. Baryon number conservation in RA is exact, not approximate. Proton decay is forbidden.*

The LLC is Lean-verified [Sandeman, 2026b]. Baryon number conservation therefore rests on machine-checked mathematics.

4.7 Maximal Parity Violation from Acyclicity

Proposition 4.3 (Maximal parity violation)**.** *The BDG action is retarded: $S_{\mathrm{BDG}}(v)$ counts intervals in* past(v) *only. The N_3/N_4 charge encodes how a pattern's causal ancestry winds around the temporal axis. To reverse this winding — to convert a left-handed pattern to right-handed — would require some causal edges to point from future to past. The DAG is acyclic: this is impossible. The weak interaction, which changes the N_3/N_4 charge of a pattern, can only couple to patterns whose winding is consistent with forward temporal propagation: the left-handed sector. Parity violation is exactly maximal — a topological impossibility, not a statistical bias [Sandeman, 2026h].*

The arrow of time and the left-handedness of the weak force are the same fact: DAG acyclicity.

4.8 Three Generations and No Fourth

The BDG action in 4D uses N_1, N_2, N_3, N_4 — four interval sizes. Persistent patterns can be excited at three levels: N_2-dominant (first generation: e, u, d, ν_e), N_3-dominant (second: μ, c, s, ν_μ), N_4-dominant (third: τ, t, b, ν_τ). There is no N_5 in the 4D BDG action. The existence of exactly three generations is a structural consequence, not a parameter to be fitted. This prediction is supported by the LEP measurement $N_\nu = 2.984 \pm 0.008$ [Particle Data Group, 2022] and by the absence of a fourth generation at the LHC.

4.9 The Koide Formula from Generation SU(3)

The three BDG excitation levels $\{N_2, N_3, N_4\}$ transform under the same permutation group S_3 as the three colour charges $\{r, g, b\}$ — both arise from the same BDG structure. Extending $S_3 \to SU(3)_{\text{gen}}$, the most general invariant charged-lepton mass matrix is:

$$M_\ell = m_0\big(I + \sqrt{2}U(\theta)\big), \quad U(\theta) = \text{diag}\big(\cos\theta, \cos(\theta + \tfrac{2\pi}{3}), \cos(\theta + \tfrac{4\pi}{3})\big). \tag{8}$$

For any m_0 and θ, the eigenvalues automatically satisfy the Koide relation [Koide, 1983]:

$$K \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \tag{9}$$

Numerically: $K = 0.666661$ with the 2022 PDG values, agreeing with $2/3$ to one part in 10^5. The fit gives $m_0 \approx 313.9$ MeV — equal to the constituent quark mass $m_p/3 \approx 312.8$ MeV to 0.35% [Sandeman, 2026h]. Both scales are set by the QCD string tension σ, connecting the lepton mass scale to the baryon mass scale through a single BDG calculation.

The Koide formula has been observed since 1981 and never derived from the Standard Model. In RA it is the generation-sector Gell-Mann-Okubo relation: a group-theoretic consequence of $SU(3)_{\text{gen}}$.

4.10 Grand Unification at the Planck Scale

$SU(3)_{\text{colour}}$ acts on the three spatial N_2 directions; $SU(3)_{\text{gen}}$ acts on the three BDG excitation levels $\{N_2, N_3, N_4\}$. Both groups arise from the same BDG structure. At the Planck scale ($\mu \to \infty$), where all BDG levels are equally and densely populated, the two $SU(3)$ groups become indistinguishable. Grand Unification is the Planck-scale limit of the BDG structure, not an additional postulate. Consequently:

$$\sin^2\theta_W\big|_{\text{GUT}} = \frac{3}{8}, \tag{10}$$

the SU(5)-type prediction, running to the observed 0.231 at the Z scale via Poisson-CSG μ-dependence. No supersymmetric partners are required: the three problems SUSY was introduced to solve (hierarchy problem, dark matter, coupling unification) are each resolved differently in RA, and the SUSY algebra is not generated by BDG dynamics.

4.11 Force Ranges from BDG Cost and LLC Balance

Force	Exchange	BDG cost	Range mechanism
EM	photon	0	$1/r^2$ from causal 2-sphere geometry
Weak	W, Z	> 0	Finite lifetime $\hbar/mc$; $\sim 10^{-18}$ m
Strong	gluon	0, but $Q_{N_2} \neq 0$	LLC flux tube; $V = \sigma r$

All three force ranges follow from the BDG actualization cost and LLC balance, with no additional parameters [Sandeman, 2026h]. Regge trajectories $m^2 \propto J$ follow directly from the linear potential. Asymptotic freedom at $\mu \gg 1$ and confinement at $\mu \to 1$ are the same Poisson-CSG μ-dependence seen at different scales.

4.12 Neutrino Masses and Oscillations

Neutrino oscillations are the generation-sector analogue of colour mixing. Each mass eigenstate $|\nu_k\rangle$ is a distinct BDG pattern with actualization frequency $f_0^k = m_k c^2/h$. Phase accumulation at different rates gives oscillation probability [Sandeman, 2026h]:

$$P(\nu_\alpha \to \nu_\beta) = \left| \sum_k U_{\alpha k}^* U_{\beta k} \exp\left(-i \frac{(m_k^2 - m_1^2) c^4 L}{2E\hbar c} \right) \right|^2 . \tag{11}$$

The CP-violating phase δ_{CP} is the same topological fact as the arrow of time: DAG acyclicity. The RA prediction: $m_1 \approx 0.37$ meV, $\sum m_\nu \approx 59$ meV, testable with CMB-S4 and Euclid.

A further structural result from the Koide fits: the lepton sector converges to $\theta_{\mathrm{lep}} = 2/9$ exactly (to one part in 10^5), where $2/9 = 2/(N_{\mathrm{gen}} \cdot N_{\mathrm{col}}) = 2/(3 \times 3)$ is determined by the same BDG integers that define $SU(3)_{\mathrm{gen}}$ and $SU(3)_{\mathrm{colour}}$. This predicts the Cabibbo angle: $|V_{us}| = \sin(2/9) = 0.2204$, agreeing with the observed 0.2245 to 1.8%. The three quark Koide angles form an arithmetic progression $\theta_0 \pm \delta$ around $\theta_0 = 2/9$, with a single displacement $\delta \approx 0.0665$ rad controlling the full CKM hierarchy (RASM Conjecture 7.3). The displacement δ satisfies $\delta = C_2(SU(2)_{\mathrm{fund}})\, \alpha_s(\mu)/\pi = (3/4)\, \alpha_s/\pi$ to 0.1% (RASM Proposition 7.4). Both the K-breaking ($\alpha_s/\pi \times 4/3$) and the θ-displacement ($\alpha_s/\pi \times 3/4$) are one-loop corrections at the same hadronic scale with their respective gauge Casimirs. Their ratio $16/9 = |c_{N_3}|/|c_{N_2}|$ is the BDG coefficient ratio. A third result follows: $|V_{cb}| = (2/\pi)\, \delta = (3\alpha_s)/(2\pi^2)$ to 0.4% (RASM Conjecture 7.4). Three CKM parameters from two RA quantities ($\theta_0 = 2/9$ and $\alpha_s(\mu)/\pi$): $|V_{us}| = \sin(2/9) = 0.2204$ (1.8%), $\delta = (3/4)\alpha_s/\pi = 0.0666$ (0.1%), $|V_{cb}| = (2/\pi)\delta = 0.0424$ (0.4%).

5 Cosmology

5.1 Dark Matter as Topology, Not Particles

Any particle that does not participate in on-shell actualization events has zero RA assembly depth and contributes zero metric weight. WIMPs, axions, and sterile neutrinos are categorically excluded as dark matter candidates — not ruled out by non-observation alone, but structurally prohibited by the actualization criterion. This is the RA WIMP prohibition [Sandeman, 2026b].

The observed dark matter phenomenology is replaced, not explained by a new particle, across four distinct lensing regimes [Sandeman, 2026g]:

- **Rotation curves:** Two RA-derived mechanisms together give flat rotation curves at all radii. In the disk ($r < r_c$), the 2D cylindrical Laplacian of $\ln \lambda$ gives $\Sigma_\lambda \propto 1/R$, yielding $v_{\mathrm{flat}}^2 = \xi/(2r_d)$ (RADM §3.1, fully derived). In the sparse halo ($\mu < 1$), the DJW theorem gives 2D gravity $\Phi \propto \ln r$, from which the Tully-Fisher relation $v_{\mathrm{flat}}^4 \propto M$ follows by causal flux conservation.
- **Bullet Cluster:** The A_{RA} ratio (stellar/shocked gas $\approx 10^8$) makes lensing trace the stellar component by a factor of 10^8 over the gas — no dark matter halo required. A quantitative $\kappa(\theta)$ profile is derived in RADM.
- **Galaxy-galaxy lensing:** The DJW logarithmic potential gives a convergence $\kappa(b) = v_{\mathrm{flat}}^2/(4G\Sigma_{\mathrm{cr}} b)$ as a theorem of the RA causal structure (RADM §4.1, RA-native derivation). This profile cuts off at the baryonic edge ($R_{\mathrm{baryon}} \sim 150$–$300$ kpc), whereas ΛCDM's NFW halo extends to ∼Mpc. The shape at $b > 300$ kpc falsifies one framework. The residual amplitude is accounted for by the warm CGM, testable via Sunyaev-Zel'dovich stacking.

- **Cosmic shear:** RA recovers ΛCDM to $\sim 1\%$ at large scales; RA-specific corrections appear at 10–100 Mpc.

5.2 The Hubble Tension

The Hubble tension — the discrepancy between the CMB-inferred value $H_0 \approx 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and the local measurement $H_0 \approx 73\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ — arises naturally in RA from the actualization density difference between matter-dominated dense regions and underdense voids. The local Universe contains the KBC supervoid, a region of anomalously low actualization density. RA predicts $H_{\mathrm{local}} \approx 73.6\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ in this void, matching the local measurement without new physics [Sandeman, 2026b,g].

A specific falsifiable prediction: the Eridanus supervoid should exhibit $H $\approx 76.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, measurable with current and near-term surveys.

5.3 The Cosmological Constant and Boltzmann Brains

Vacuum energy suppression is a direct consequence of the $\mathscr{P}_{\mathrm{act}}$ operator: only on-shell actualized states contribute to the metric source. Virtual fluctuations are excluded, resolving the cosmological constant problem without fine tuning [Sandeman, 2026b].

The Boltzmann Brain problem is categorically resolved by two mechanisms. First, *structurally*: each irreversible actualization requires a thermodynamic cost that cannot be met by vacuum fluctuation—a brain requires $\sim 10^{25}$ independent actualization steps, each crossing the ΔS^* threshold. Second, *dynamically*: the dual severance mechanism (RAGC §6.6) fragments every causally connected component on a finite timescale, so no patch persists long enough for BB nucleation. The combination is a categorical prohibition, not merely a probabilistic suppression [Sandeman, 2026b].

6 Complexity and Life

6.1 The Causal Firewall

At the critical density $\mu = 1$, the Poisson-CSG undergoes an Erdős-Rényi giant-component transition. Below this threshold, causal graphs are locally tree-like — chemistry. Above it, they develop stable long-range causal connectivity — life. The Causal Firewall is the RA account of the origin of life: not a continuum but a percolation threshold, as sharp as a phase transition [Sandeman, 2026e].

This is the same $\mu = 1$ transition as QCD confinement, galactic rotation curve onset, and quantum fault-tolerance threshold. One mathematical transition, four physical manifestations.

6.2 Assembly Theory Convergence

Assembly Theory [Walker et al., 2023] independently arrived at the concept of causal depth as the measure of biological complexity. In RA, causal depth $A_{\mathrm{RA}}(v)$ — the length of the longest directed path to vertex v — is the fundamental quantity. The convergence of two independent frameworks on the same mathematical structure supports both.

New results in RAHC and RACI [Sandeman, 2026d,e] extend this to metabolic networks via the *coarse-grained assembly depth* $\tilde{A}_{\mathrm{RA}}(\mathscr{N})$: the number of irreversible reactions on the critical path through the network. For glycolysis $\tilde{A}_{\mathrm{RA}} = 3$ (hexokinase, PFK-1, pyruvate kinase), with allosteric regulation concentrating at exactly these three nodes — independent biological confirmation. The *origin-of-life sandwich* $A(M_0^*) \leq \tilde{A}_{\mathrm{RA}}(\mathscr{N}_0) \leq \tau_{\mathrm{cycle},0}/\tau_{\mathrm{irrev}}$ brackets the causal depth of the first replicators from both sides.

Causal depth is a selection pressure: replication rate is exponentially suppressed by $A_{\mathrm{RA}}(M)$ (England bound), so natural selection for replication speed is simultaneously selection for minimal

causal depth. Intelligence is recursive causal closure: the capacity of a system to model its own actualization history and use that model to guide future actualization [Sandeman, 2026d].

6.3 Substrate-Independent Habitability (RACF)

The Causal Firewall criterion is substrate-independent: neither Condition F1 (Quantum Darwinism redundancy $R \geq R_{\min}$) nor Condition F2 (Unruh temperature $T_U = \hbar a/2\pi c k_B \ll T_{\text{env}}$) makes reference to molecular class, solvent chemistry, or biological alphabet. The circumstellar habitable zone is a terrestrial proxy for these deeper physical conditions, not the physical criterion itself. RACF [Sandeman, 2026e2] develops three universal biosignature observables and two falsifiable predictions for the distribution of life, submitted to *International Journal of Astrobiology*.

7 Quantum Information

The Kinematic Coherence Bound [Sandeman, 2026f] establishes that the maximum number of fault-tolerant logical qubits in a physical array scales as $N_{\max} = \eta \cdot p_{\text{th}}$, where $p_{\text{th}} \approx 10^{-2}$ is the per-link actualization probability. This threshold is not a property of any specific error-correction code: it is the $\mu = 1$ Erdős-Rényi transition applied to the causal neighbourhood of the logical qubit.

The Lean-verified BDG particle classification (RATM [Sandeman, 2026RATM]) establishes a structural account of qubit fragility. Every stable free particle has a positive BDG actualization score; the minimum is 1. Electrons and photons — the particles that carry quantum information in all current hardware — propagate at the sequential fixed point with score 1. $W^{\pm}$ bosons score 9; gluons score 18. Qubit fragility is therefore not an accident of electromagnetic coupling strengths: it is a consequence of the fact that laboratory-accessible particles sit at the minimum BDG stability that the causal filter permits. This result is proved by `D1a_fixed_point` in `RA_D1_Proofs.lean`.

The BDG particle universe closure theorem (RATM, Theorem D1_closure_complete, 124 extension cases exhaustively verified) additionally constrains the decoherence channels available to any qubit: every environmental particle that can couple to a quantum system belongs to the closed BDG set. No exotic decoherence mechanism from outside the closure is possible within the RA framework.

The Lieb-Robinson bound on actualization propagation velocity $v_{\text{LR}} \sim J\ell/\hbar$ is the lattice expression of the same causal bandwidth ceiling as $c = l_P/t_P$ in the relativistic regime: the maximum rate at which actualization can propagate through the causal graph, at the relevant density scale.

The fault-tolerance threshold, the speed of light, the Causal Firewall, and the QCD confinement transition are the same Erdős-Rényi percolation at four different scales of the causal graph.

8 Quantum Mechanics: Exact at the Discrete Level

8.1 The Measurement Problem Dissolved

The measurement problem asks: why does a quantum superposition resolve into a definite outcome when measured? In RA the question does not arise, because the premise is wrong. There is no superposition that mysteriously collapses; there is a causal graph that grows. A quantum measurement is an actualization event at which the entropic criterion $\Delta S(\rho \| \sigma_0) > 0$ is satisfied. At that vertex, one outcome becomes permanently encoded as a directed edge in the DAG. The other outcomes are the space of *unactualized candidates* — they were never in the graph; they were possible extensions of the graph that were not selected.

Wave function collapse is the writing of a new vertex. The wave function is not a physical object that collapses; it is a description of the probability distribution over possible next actualization events.

8.2 The Quantum Measure is Exact; the Schrödinger Equation is Not

The quantum measure of Sorkin [Sorkin, 1994], assigning complex amplitudes to causal histories in a foliation-independent way, is exact at the discrete level. It is not an approximation to RA: it IS the RA causal graph dynamics at Level 2. Quantum mechanics, in the sense of the full quantum measure formalism, is therefore not a macroscopic limit — it is the primitive description of the growing DAG.

What is a macroscopic approximation is the *Schrödinger equation* specifically. It emerges in the slowly varying, large-μ, non-relativistic, single-particle continuum limit of the quantum measure:

$$\mu_{\mathscr{C}}(\text{transition: } |i\rangle \to |f\rangle) \xrightarrow{\mu \gg 1} \left| \int \mathscr{D}\Phi \, e^{iS[\Phi]/\hbar} \right|^2 \xrightarrow{\text{non-rel.}} \text{Schrödinger evolution.} \tag{12}$$

The hierarchy is: quantum measure (exact) $\supset$ Feynman path integral (large-μ limit) $\supset$ Schrödinger equation (non-relativistic single-particle limit). Heisenberg uncertainty is the discrete graph's resistance to simultaneous specification of complementary causal properties — exact at the discrete level, not an approximation.

8.3 Classical Behaviour as the Dense-Graph Regime

A large, warm, complex system is not fundamentally different from a quantum system — it IS a quantum system, described by the same discrete causal graph. The difference is density. In the regime $\mu \gg 1$, every degree of freedom has been pinned by previous actualization events: the causal graph is so densely connected that quantum indeterminacy has been eliminated at every relevant scale. Classical behaviour is not a different ontology from quantum behaviour; it is the $\mu \gg 1$ regime of the same ontology. The transition from quantum to classical is the gradient $\mu(x)$ increasing from sparse to dense. There is no Heisenberg cut; there is only a gradient in actualization density.

The famous paradoxes of quantum mechanics — the double slit, the delayed choice, the quantum eraser, Schrödinger's cat — are each resolved by identifying where in the growing DAG the actualization events occur and what constraints they impose on the remaining candidates [Sandeman, 2026a].

9 The Experimental Programme

Relational Actualism makes a large number of specific, quantitative, falsifiable predictions. Table 1 collects the mass predictions that follow from the cascade formula $m_p = m_P \, \alpha_{\text{EM}}^5 / 2^{28}$ (RASM Derivation 2). Table 2 lists further predictions.

Table 1: Mass and radius predictions from the μ^5 cascade (RASM Derivation 2). Inputs: m_P, α_{EM} (IC30), $L_q = 4$ (BDG, Lean-verified), $c_4 = 8$ (BDG).

Quantity	Formula	Predicted	Observed	Error
m_p	$m_P \, \alpha_{\text{EM}}^5 / 2^{28}$	941 MeV	938 MeV	0.3%
m_n	$= m_p$ (leading)	941 MeV	940 MeV	0.2%
r_p	$L_q \times l_C$	0.841 fm	0.841 fm	0.03%
m_π	$(2/3)^5 \, m_p$	124 MeV	140 MeV	11%
m_W	$6^{5/2} \, m_p$	83 GeV	80 GeV	3.3%
m_H	$(\alpha_{\text{EM}}^{-1} - L_q) \, m_p$	125.2 GeV	125.3 GeV	0.06%

The most decisive near-term test is the Bose-Marletto-Vedral (BMV) protocol [Bose et al., 2017, Marletto & Vedral, 2017]: two masses placed in spatial superposition, entangled gravitationally. If gravity-mediated entanglement is observed, the fundamental premise of RA — that the spacetime metric is updated strictly from actualized vertices — is falsified. RA predicts a strict null result.

The most important medium-term test is the neutrino mass sum. Several forthcoming experiments (CMB-S4, Euclid, DESI) will reach sensitivity $\sum m_\nu \sim 20$ meV. RA predicts $\sum m_\nu \approx 59$ meV. If the bound falls below this, the $SU(3)_{\text{gen}}$ structure requires revision.

10 Open Derivations

The programme originally identified four derivation targets. All four have been resolved or dissolved. None remain open.

D1: Coupling constants — resolved. The BDG integers $(1,-1,9,-16,8)$ are the *unique* growth rule for a self-consistent $d=4$ causal graph: the O14 uniqueness theorem establishes that binomial inversion (Möbius inversion on the Boolean lattice) applied to the $d=4$ chord diagram moment sequence $r_k = (2k+3)(2k+2)(2k+1)/6$ (Yeats 2025) yields exactly these integers with zero free parameters (Rota 1964). The coupling constants follow: $\alpha_{\text{EM}}^{-1} = 137$ (Lean-verified, L08) with dressed value 137.036 via the discrete Dyson equation (IC30, 0.00001% of PDG); $\alpha_s(m_Z) = 1/\sqrt{72} = 0.11785$ (0.13% of PDG). No MCMC simulation is needed — the coupling constants are uniquely determined by the BDG integer structure.

D2: Gauge group structure — resolved. The Standard Model gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ emerges from the alternating sign pattern of the BDG coefficients $(+,-,+,-,+)$, which is forced by inclusion–exclusion counting of causal intervals (GS02, confirmed by exact enumeration [CV]; see RATM §7.6). The $2+1$ split at depth 1 ($E[N_1]=1$ at $\mu=1$, selecting one spatial direction) produces $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; the depth-3 and depth-4 sectors produce $\mathrm{SU}(3)_c$. The remaining open step is deriving the $\mathrm{SU}(3)_{\text{gen}}$ breaking pattern that explains why Koide holds exactly for leptons but approximately for quarks.

D3: Lean formal proofs — resolved. The BDG particle classification, colour confinement, and closure theorem were already Lean-verified (`RA_D1_Proofs.lean`). The chirality theorem (maximal parity violation from DAG acyclicity + BDG filter) and baryon conservation (N_2 preserved in all asymptotic extensions, verified exhaustively) are now proved in `RA_BaryonChirality.lean` (13 theorems, zero sorry tags, compiled April 2026). The weak force is left-handed because time has a direction: DAG acyclicity forbids backward-pointing edges, and the BDG integers make the backward-winding pattern $N=(1,1,1,0)$ unstable (score $=-7<0$).

D4: Type III_1 AQFT extension — dissolved. The finite-dimensional Lean proofs of frame independence and Rindler stationarity are the fundamental results in RA's discrete ontology. The type III_1 von Neumann algebra framework (Tomita-Takesaki modular theory) is the continuum approximation to these discrete facts—it is what the Lean-verified results *look like* when coarse-grained into continuum AQFT at $\mu \gg 1$. Showing compatibility is a consistency check with the same dissolved status as O10 (Bianchi = LLC in continuum costume) and O11 (Lorentz = causal invariance in continuum costume). It is not a foundational requirement for the framework.

The history of the programme is that each target has yielded to an RA-native insight once stated precisely in the framework's own terms. D4 is the latest example: the "problem" was asking the discrete theory to prove compatibility with a continuum approximation, which inverts the ontological priority.

11 Further Consequences

Four additional long-standing problems dissolve immediately once stated in RA-native terms. Full derivations appear in RASM [Sandeman, 2026h].

The strong CP problem: $\theta_{\text{QCD}} = 0$ exactly. The QCD vacuum angle θ_{QCD} is generated in the Standard Model by instantons — field configurations that tunnel between topological vacuum sectors. Instantons require closed loops in Euclidean spacetime. In RA, time is the direction of causal edges; a closed Euclidean loop requires a closed causal loop. The DAG is acyclic: instantons do not exist. $\theta_{\text{QCD}} = 0$ exactly, not merely small. A second independent argument: the LLC on N_2 spatial charge is symmetric under charge conjugation by construction, making CP an exact symmetry of the strong-force LLC. The axion was introduced to relax θ_{QCD} to zero dynamically. RA predicts the axion does not exist — consistent with all observations.

The black hole information paradox: dissolved. A black hole is a Causal Severance: a topological partition of the causal graph at which outgoing edges are absent. The LLC holds independently on each side. Information is not lost; it is partitioned — encoded in the initial LLC conditions of the daughter spacetime. An exterior observer traces over the interior degrees of freedom and sees thermal (Hawking) radiation, not because information is destroyed but because causal disconnection produces a mixed state for any single-component observer. The total system is unitary. The paradox arises only if one incorrectly requires a single exterior observer to access causally disconnected information — which Causal Severance structurally forbids.

The fine structure constant as self-consistency eigenvalue (conjecture). The per-link actualization probability satisfies $p_{\text{th}} = \alpha_{\text{EM}}$ via the optical theorem (proved, RASM): the fraction of EM interaction candidates satisfying the actualization criterion $\Delta S > 0$ equals α_{EM}. Combined with the score-1 floor theorem (RATM, Lean-proved), this gives the self-consistency condition $\mu_{\text{stable}} = 1/\alpha_{\text{EM}} \approx 137$.

The empirical Wyler formula [Wyler, 1969] $\alpha = 9/(8\pi^4)(\pi^5/1920)^{1/4}$ (accurate to 0.00006%) decomposes with all factors BDG-derived: $|c_{N_2}| = 9$, $d = 4$, and $\Gamma(5/4)$ from the causal diamond volume integral — the same $\Gamma(5/4)$ that appears in the BDG locality lemma (RACL). Two of the three conditions needed to derive α from BDG integers alone are established; the third (computing $V(SU(3)_{\text{gen}})$ in BDG normalisation) is the remaining open step (RASM Conjecture 7.2).

The Cabibbo angle from $SU(3)_{\text{gen}}$ (conjecture). The Koide fit converges to $\theta_{\text{lep}} = 2/9$ to one part in 10^5, where $2/9 = 2/(N_{\text{gen}} \cdot N_{\text{col}}) = 2/(3 \times 3)$. This predicts the Cabibbo angle: $|V_{us}| = \sin(2/9) = 0.2204$, compared to the observed 0.2245 (1.8% error). The three quark Koide angles form an arithmetic progression $\theta_0 \pm \delta$ around $\theta_0 = 2/9$, with $\delta \approx 0.0665$ rad controlling the full CKM hierarchy. Two numbers — $2/9$ and δ — parametrize all three CKM mixing angles. (RASM Conjecture 7.3.)

Why $d = 4$: four independent uniqueness arguments. The BDG action in d dimensions has $d/2 + 1$ non-gravitational degrees of freedom. Argument 1: only $d = 4$ gives exactly three gauge forces. Argument 2 (D4U02): the selectivity ceiling $\Delta S^*(\mu)$ has its maximum at $\mu \approx 1$ only in $d = 4$. Argument 3 (new): the cascade exponent $2N_c - 1$ equals the confinement cycle length $d + 1$ only in $d = 4$. Argument 4 (new): the Alexandrov/sphere geometric factor $(4/3)\,d!$ equals the BDG confinement parameter $d \times 2^{d-1}$ only in $d = 4$; this is what allows hadronic self-consistency (RASM Proposition **??**). Stable composite matter — protons, neutrons, atoms — can only exist in $d = 4$.

The hierarchy problem: dissolved. The proton mass $m_p \approx 10^{-20} m_P$ is not "unnaturally small." It is the *unique* mass satisfying the BDG self-consistency condition in $d = 4$: the cascade formula $m_p = m_P \alpha_{\mathrm{EM}}^5 / 2^{28}$ follows from the gluon vertex probability at the self-consistent internal density $\mu_p = \alpha_{\mathrm{EM}}/32$, with $N_{\mathrm{eff}} = L_q^3$ derived from the $d = 4$ geometric identity. There is no fine-tuning because there is no free parameter to tune; the BDG integers fix everything.

The Higgs mechanism in RA. In RA, the "Higgs field" is the background depth-2 actualization density. The W/Z bosons are marginal BDG vertices with score $S(1,0,0,0) = c_0 + c_1 = 0$ exactly; they require depth-2 dressing to actualize. The photon escapes as $N = (1,1,0,0)$ with $S = 9$, actualizing freely. The Higgs boson is a coherent perturbation of 133 background modes, with mass $m_H = 133 m_p = 125.2$ GeV (RASM equation **??**). No fundamental scalar field is needed.

The baryon asymmetry as a Causal Severance initial condition. Baryon number is exactly conserved in RA (Proposition 5.1 in RASM). Sakharov's first condition — baryon number violation — is therefore absent. This does not prevent RA from accounting for η; it means dynamic baryogenesis does not occur. The Big Bang is a Causal Severance nucleation: our universe inherited its net baryon number $B \neq 0$ from the LLC boundary conditions at that event. The baryon asymmetry $\eta \approx 6.1 \times 10^{-10}$ [Particle Data Group, 2022] is an initial condition set at the Planck scale, not a dynamic outcome. The prediction: no signatures of electroweak baryogenesis in CMB data; the baryon-to-photon ratio is isotropic at all scales.

12 Lean 4 Formal Verification

The RA programme uses Lean 4/Mathlib as a proof assistant to machine-verify the discrete graph-theoretic core of the framework. This is not cosmetic: the Lean proofs establish that the claims about the growing DAG are internally consistent at the level of formal mathematics, independently of any physical interpretation.

The current Lean corpus comprises **176 verified theorems across seven proof files**, with one intentional sorry (the LQI adapter in `RA_AQFT_Proofs_v10.lean`) and no axioms beyond Mathlib:

Result	File	Status
Local Ledger Condition (L01)	`RA_GraphCore.lean`	Proved, 0 sorry
Graph Cut Theorem (L02)	`RA_GraphCore.lean`	Proved, 0 sorry
Markov Blanket Shielding (L03)	`RA_GraphCore.lean`	Proved, 0 sorry
Amplitude locality (O01)	`RA_AmpLocality.lean`	Proved, 0 sorry
Causal invariance (O02)	`RA_AmpLocality.lean`	Proved, 0 sorry
Frame independence of $\Delta S(\rho\|\sigma_0)$	`RA_AQFT_Proofs_v10.lean`	Proved, 0 sorry
Rindler stationarity (Unruh resolution)	`RA_AQFT_Proofs_v10.lean`	Proved, 0 sorry
Koide $K = 2/3$ identity (L09)	`RA_Koide.lean`	Proved, 0 sorry
$SU(3)_{\text{gen}}$ coherent state theorem	`RA_D1_Proofs.lean`	Proved, 0 sorry
BDG particle classification (5 types, L11)	`RA_D1_Proofs.lean`	Proved, 0 sorry
BDG closure theorem (124 cases)	`RA_D1_Proofs.lean`	Proved, 0 sorry
Colour confinement (L=3, L=4; L10)	`RA_D1_Proofs.lean`	Proved, 0 sorry
$\alpha_{\text{EM}}^{-1} = 137$ (L08)	`RA_D1_Proofs.lean`	Proved, 0 sorry
$\mathscr{P}_{\text{act}}$ conservation theorem	`RA_D1_Proofs.lean`	Structural proof
BDG locality lemma	numerically verified	Confirmed

Amplitude locality (O01) is proved as a theorem of discrete DAG dynamics—the BDG action increment depends only on the causal past by transitivity of the causal order—making causal invariance (O02) unconditional with zero sorry tags. The type III_1 AQFT framework (Tomita-Takesaki modular theory, von Neumann algebras) is the continuum approximation to these discrete results; in RA's ontology, the finite-dimensional Lean proofs are fundamental and compatibility with the continuum is a consistency check, not a foundational requirement (same dissolved status as O10 and O11).

A Python notebook independently confirms 52/52 numerical checks at machine precision, covering the Koide formula, the BDG coefficient ratios, the Wyler formula decomposition, and the coupling constant hierarchies. The machine verification strategy — using Lean for structural results and Python for numerical confirmation — provides two independent lines of evidence for the quantitative claims of the suite.

13 Prior Approaches and the Position of RA

General relativity and quantum field theory. Both frameworks are derived in RA as macroscopic approximations to the causal graph dynamics. They are not competing descriptions of nature; they are Level 3 descriptions that apply in different density regimes of the same Level 1 algorithm.

Loop Quantum Gravity. LQG derives geometry from a spin network background but imports the Standard Model as separate matter content. It offers no account of why the gauge group has three factors or why charge is quantised. RA requires no background; the Standard Model structure is a consequence of 4D BDG topology.

String Theory. String theory can reproduce a rich particle spectrum but requires 10 or 11 spacetime dimensions and a landscape of $\sim 10^{500}$ vacua with no selection principle. RA is uniquely 4-

dimensional by BDG + Lovelock, has no landscape, and no free parameters at the topological level.

Causal set theory. RA is built directly on the causal set foundation of Sorkin [Sorkin, 1994] and Bombelli [Bombelli et al., 1987], extending it with three new elements: (1) the BDG actualization filter $S_{\mathrm{BDG}}(v) > 0$ as a physical selection criterion (rather than a dynamics to be inferred from path integrals); (2) the LLC as a vertex-level conservation law, Lean-verified; (3) the Poisson-CSG growth model as the mean-field description of the stochastic graph rewriting. The key advance over standard causal set theory is the BDG locality lemma (RACL): the dense-graph limit of the BDG sum is provably local, which is what allows Lovelock's uniqueness argument to apply. Without locality, Lovelock does not constrain the field equation; with it, Einstein is forced. Causal set theory asks: can GR emerge from a causal set? RA answers: it must, and here is a machine-checkable proof.

The Rideout-Sorkin Classical Sequential Growth model [Rideout & Sorkin, 2000] is the direct precursor to the RA Poisson-CSG: the CSG dynamics supply the Born-rule-like probability measure for graph growth that RA uses. RA extends CSG by adding the BDG filter and identifying the LLC as the conservation law that forces GR.

Bilson-Thompson topological preons. Bilson-Thompson's braid model [Bilson-Thompson, 2005] maps the first-generation Standard Model onto ribbon braids. RA is in the same direction but derives the stability selection rule (the BDG filter) that the braid model lacks. Whether RA's stable patterns correspond to Bilson-Thompson braids is an open question that D1 would resolve.

14 Conclusion

Physics has spent a century seeking the reconciliation of quantum mechanics and general relativity, and asking why the Standard Model has the structure it has. Relational Actualism proposes that both questions are symptoms of a shared error: treating spacetime, matter, and quantum states as primitive, when none of them is.

The primitive is the actualization event. Not the particle, not the field, not the wave function, not the spacetime manifold. The moment at which a physical interaction becomes irreversible — at which $\Delta S(\rho \| \sigma_0) > 0$ and a permanent causal edge is written into the DAG.

From this:

Result	Status
$c = l_P/t_P$; $E = \gamma mc^2$; $\tau = N_{\text{act}} \cdot t_P$	Proved
GR inevitable (BDG uniqueness + Lovelock)	Proved
Three non-gravitational forces in 4D	Proved
Charge quantisation in units of $e/3$	Proved
Photon masslessness	Proved
Colour confinement	Proved
Exact baryon conservation	Proved (Lean-backed)
Maximal parity violation	Proved
Exactly three generations	Proved
Koide lepton mass formula	Conjectured; $K = 2/3$ exact
GUT unification at Planck scale	Conjecture
Flat rotation curves; Hubble tension	Derived
$\kappa(\theta)$ Bullet Cluster profile	Qualitative (RADM §4.2)
4-regime lensing classification	Established (RADM §4.3)
DJW lensing $\kappa(b) = v_{\text{flat}}^2/(4G\Sigma_{\text{cr}}b)$	Derived (RADM §4.1)
$\kappa(b)$ cutoff at R_{baryon} vs NFW	Falsifiable prediction
Galaxy-galaxy amplitude (CGM)	Prediction, open calc
Origin of life (Causal Firewall)	Derived
QFT fault-tolerance threshold	Derived
Measurement problem	Dissolved
$\theta_{\text{QCD}} = 0$ exactly; no axion	Proved
Black hole information paradox	Dissolved
Baryon asymmetry as Causal Severance IC	Derived
$d = 4$ spacetime dimensionality	Conjecture
Quark Koide breaking (RA form; α_s external input)	Conjecture
Wyler formula for α_{EM} (0.00006%)	Conjecture
$\theta_{\text{lep}} = 2/9$ (Koide fit, $1:10^5$)	Conjecture
$\lvert V_{us}\rvert = \sin(2/9)$ Cabibbo prediction (1.8%)	Conjecture
$\delta = \frac{3}{4}\alpha_s/\pi$ (0.1%)	Proved
$\lvert V_{cb}\rvert = (2/\pi)\delta$ (0.4%)	Conjecture

The question driving the next phase of the programme is exact: is the complete set of $S_{\text{BDG}} > 0$-stable self-similar patterns in a 4D BDG-filtered Poisson-CSG precisely the first-generation Standard Model particle spectrum? If yes — if the BDG-filter MCMC confirms that the stable patterns are exactly $\{e, u, d, \nu_e, \gamma, g, W, Z, H\}$ — then all of physics follows from the single fact that some things happen and others do not.

The programme as it stands in early 2026:

- RAQM at *Foundations of Physics* (awaiting editor assignment). RACL and RATM targeting *Physical Review D* (revised after initial rejections). RACF at *International Journal of Astrobiology* (under review). RACL targeting *Physical Review D* (revised after CQG desk rejection).

- Twelve papers in the suite; eleven on Zenodo with permanent DOIs (RACF submitted to *International Journal of Astrobiology*).
- 176 Lean-verified theorems across eight proof files, one intentional sorry (LQI adapter).
- 52/52 independent numerical checks at machine precision.
- All four derivation targets resolved or dissolved: D1 (coupling constants), D2 (gauge groups), D3 (chirality/baryon, compiled April 2026), D4 (type III_1 AQFT, dissolved).

The programme does not claim completeness. It claims internal consistency at the discrete level (machine-verified), correct macroscopic limits (GR inevitable, Standard Model derived), and specific falsifiable predictions (BMV null result; no WIMP; $\sum m_\nu \approx 59$ meV; $H_{\mathrm{Eridanus}} \approx 76.8$ km/s/Mpc; $\kappa(b)$ cutoff at R_{baryon}). The framework is falsifiable. The search continues.

Declarations

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Ethics statement: This article does not contain any studies with human participants or animals performed by the author.

AI use disclosure: Claude (Anthropic, model `claude-sonnet-4-6`) was used in the preparation of this manuscript, including drafting and revision of text, mathematical derivation assistance, LaTeX compilation, and Lean 4 proof development. The author is solely responsible for all scientific content, mathematical claims, and conclusions. Gemini (Google DeepMind, Gemini 2.0 Pro) was used for supplementary literature checking and cross-verification of symbolic calculations.

Data availability: All Lean 4 proof files and Python verification scripts are publicly available at github.com/jsandeman/Relational-Actualism.

Preprint availability: doi.org/10.5281/zenodo.19229438.

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Table 2: Falsifiable predictions of Relational Actualism.

Prediction	Value / Status	Timescale
BMV gravity-mediated entanglement	Strict null result	Near-term
No WIMP above neutrino floor; lensing from CGM halos	WIMP prohibited structurally; 4 lensing regimes	Ongoing
No fourth generation	N_5 absent in 4D BDG	LHC ongoing
No proton decay	LLC exact	Ongoing
Eridanus supervoid H	$\approx 76.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	~ 5 years
Hubble tension parameter-free	$H = H_{\mathrm{CMB}}(1 + 0.463\lvert\delta\rvert)$	DESI
Neutrino mass sum	$\sum m_\nu \approx 59$ meV	CMB-S4/Euclid
$\sin^2\theta_W$ at GUT scale	$3/8$	Existing data
$d = 4$ unique (four arguments)	Proposition	RASM
Stable matter only in $d = 4$	From $(4/3)d! = d \times 2^{d-1}$	RASM
$\theta_{\mathrm{QCD}} = 0$ exactly; no axion	Proposition	RASM
$\lvert V_{us}\rvert = \sin(2/9) = 0.2204$	Conjecture (1.8% error)	RASM Conj. 7.3
GW dispersion at Planck scale	Discrete corrections	Future GW detectors
CMB axis-of-evil persists in CMB-S4	$> 3\sigma$; structural, not statistical	CMB-S4
CMB $(2/\ell)^2$ multipole hierarchy	Decreasing alignment for $\ell \geq 4$	CMB-S4
Parent SMBH mass	≈ 2.5–$6.6 \times 10^6\, M_\odot$	RAGC §6.3
Near-minimum inflation	$N_{\mathrm{total}} \approx 64$–$65$	Tensor-to-scalar ratio
Cosmic web–CMB axis correlation	Filaments aligned with spin axis	DESI/Euclid
No heat death	Fragmentation via dual severance	RAGC §6.6
Boltzmann Brains categorically impossible	Structural + dynamical	RAGC §6.7
$\alpha_{\mathrm{EM}}^{-1} = 137.036$	Discrete Dyson equation, 0.00001%	RASM